

Copper Cables — Guided Notes

Unit 1.3.1 | CompTIA Network+ N10-009 Obj. 1.3 | N10-008 1.3

Name: _____ Date: _____ Period: _____

Complete each question using the lesson reading. Write in the blank or answer in the space provided.

Question 1

The standard Ethernet cable used in most LANs is _____ (UTP/STP). The engineering trick that cancels out EMI inside this cable is the _____ of the wire pairs.

Question 2

Match each cable category to its maximum speed at 100 meters:

Cat5e → _____

Cat6 → _____ (at 100m) / _____ (at 55m)

Cat6a → _____

Question 3

Cat8 supports speeds of _____ Gbps but is limited to _____ meters. It is designed for _____ environments, not workstation drops.

Question 4

Coaxial cable carries broadband internet from the ISP to the cable modem using the _____ standard. In data centers, a two-conductor variant called _____ cable is used for short high-speed switch connections.

Question 5

List the three jacket ratings from most to least restrictive and describe where each must be used:

1. _____
2. _____
3. _____

Question 6

A straight-through cable connects _____ devices (e.g., PC to switch). A crossover cable connects _____ devices (e.g., switch to switch). A rollover/console cable reverses all _____ pins and is used to connect a laptop to a device's _____ port.

Question 7

The maximum copper Ethernet segment length is _____ meters total. The TIA/EIA standard budgets _____ meters for the permanent link and _____ meters for patch cables. If a run exceeds this limit, the two options are _____.

Question 8 — Real World Application

REAL WORLD: A user reports their workstation connects at 100 Mbps instead of 1 Gbps. The cable run is 85 meters and passes near fluorescent light fixtures for about 20 feet. What are the two most likely causes and how would you confirm each?

Download the answer key from the class server:

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wget http://[SERVER]/resources/network-engineer/1-3-1-copper-cables/1-3-1-copper-cables-answer-key.pdf
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